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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 2	3 May 2023
Time : 1.5 hours	Maximum marks: 52
	Full marks: 50

- This question paper consists of 11 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	TOTAL
8	5	6	2	5	8	8	5	5	52

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(f) Does $A\mathbf{x} = A\mathbf{y}$ (for this matrix A) imply that $\mathbf{x} = \mathbf{y}$? Justify your answer.

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Question 2. [5 points] Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of all 2×2 real matrices, with the usual addition and scalar multiplication for matrices, and let

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \right\}.$$

(a) Is S a basis for $M_{2 \times 2}$? Justify your answer.

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(b) Find a basis for $\text{span}(S)$.

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- (c) An incomplete description of $\text{span}(S)$ is given below; there should be an equation in the place of the dots.

$$\text{span}(S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : \dots \right\}$$

Write down this equation.

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Question 3. [6 points] Let

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 3 & -2 & -3 \\ 0 & 0 & -2 \end{pmatrix}.$$

- (a) Find $\det(A)$.

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- (b) Find all the eigenvalues of A .

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(c) Find all the eigenvectors corresponding to the largest eigenvalue of A .

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Question 4. [2 points] Let B be a skew-symmetric $n \times n$ matrix, where n is odd. Show that $\det(B) = 0$.

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Question 5. [5 points] Let

$$p_1(x) = x + 1, \quad p_2(x) = x^2, \quad p_3(x) = x^3 + x$$

$$p_4(x) = x^3 + x^2 - 1, \quad p_5(x) = x^3 - 1, \quad p_6(x) = x^3 - x$$

and $A = \{p_1(x), p_2(x), p_3(x)\}$, $B = \{p_1(x), p_3(x), p_5(x)\}$.

From the list choose, if possible, a polynomial $p_i(x)$ ($1 \leq i \leq 6$) such that:

(a) $A \cup \{p_i(x)\}$ is a basis for P_3 .

(b) $B \cup \{p_i(x)\}$ is a basis for P_3 .

(c) $A \cup \{p_i(x)\}$ is linearly dependent and $B \setminus \{p_i(x)\}$ is a basis for $\text{span}(B)$.

Write **only one of** $p_1(x), p_2(x), p_3(x), p_4(x), p_5(x), p_6(x)$ or “Impossible” in the appropriate block. You may use a polynomial more than once, or not at all.

Question 6. [8 points] Read the following proof and then answer the questions following it:

Proof: Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ a subset of V such that $\text{span}(S) = V$. Suppose S is linearly dependent. (1)

Then there is an i such that $\mathbf{v}_i$ can be written as a linear combination of the vectors in $S \setminus \{\mathbf{v}_i\}$ (2)

It follows that $V = \text{span}(S \setminus \{\mathbf{v}_i\})$ (3)

But this gives a contradiction. (4)

(a) State the result proved here.

(b) Explain how the statement at (3) follows from (2).

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(c) What is the contradiction referred to at (4)?

(d) Suppose R is a subset of V such that $\text{span}(R) = V$, but that R is not a basis of V . For each of the following statements, either prove it, or disprove it by giving a counter example:

(i) R has more than n elements.

(ii) Every subset of R with n elements is a basis for V .

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Question 7. [8 points] Let the function $T : P_2 \rightarrow P_3$ be defined by $T(p(x)) = xp(x) - x^2p'(x)$, where as usual p' denotes the derivative of p .

(a) Find $T(p(x))$ when $p(x) = ax^2 + bx + c$.

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(b) Show that T is a linear transformation.

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(c) Find $\ker(T)$. Show your working.

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(d) Is T one-to-one? Give a reason for your answer.

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(e) Give, if possible, an example of an element of P_3 which is not in the range of T .

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Question 8. [5 points] Let $A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ and V be the vector space of all 2×2 matrices.

Define

$$S = \{B \in V : BA = AB\}.$$

Answer the following questions, giving reasons for all your answers.

(a) Is S closed under scalar multiplication?

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(b) Is S closed under addition?

(c) Is S a subspace of V ?

Question 9. [5 points]

(a) Find a matrix A such that $CS(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$ and $RS(A) = \text{span}\{(1, 0, 1), (1, 2, 0)\}$.

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(b) Why is it not possible to find a matrix A such that $NS(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$ and $RS(A) = \text{span}\{(1, 0, 1), (1, 2, 0)\}$?